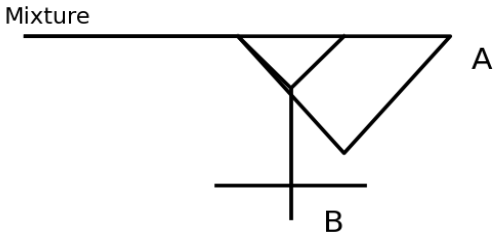
20 JAMB UTME CHEMISTRY

Diagram-Based Questions with Answers and Explanations

20 original UTME-style multiple-choice practice questions

Instructions: Study each diagram, select the correct option, then read the explanation immediately below the options. These are original practice questions, not verbatim official JAMB past-paper items.

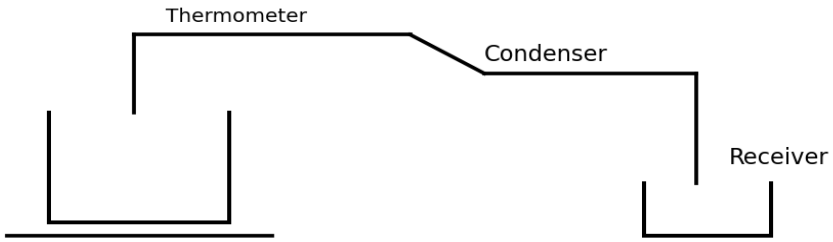
1. The apparatus shown is used to separate an insoluble solid from a liquid. What are A and B respectively?



A. Residue and filtrate
B. Filtrate and residue
C. Solvent and solute
D. Distillate and residue

Correct answer: A
Explanation: The solid trapped on the filter paper is the residue (A); the liquid that passes through is the filtrate (B).

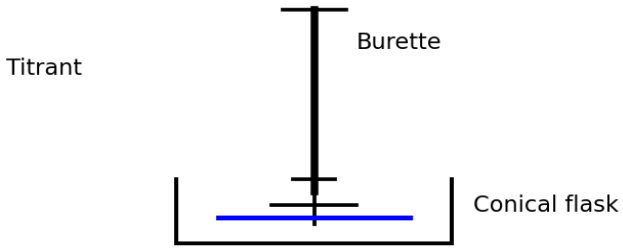
2. In the distillation setup shown, the main function of the condenser is to



A. heat the mixture
B. cool vapour into liquid
C. increase the boiling point
D. remove dissolved salts

Correct answer: B
Explanation: Cold water removes heat from the vapour, causing it to condense into liquid distillate.

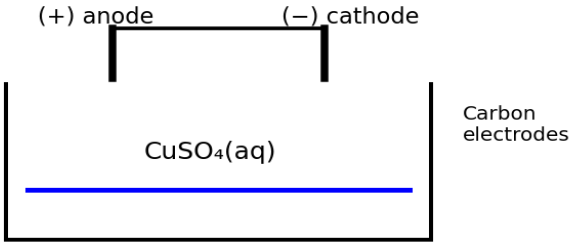
3. During an acid–base titration, the liquid placed in the burette is usually the



A. indicator only
B. unknown solid
C. standard solution or titrant
D. wash water

Correct answer: C
Explanation: The burette delivers a measured volume of titrant into the conical flask.

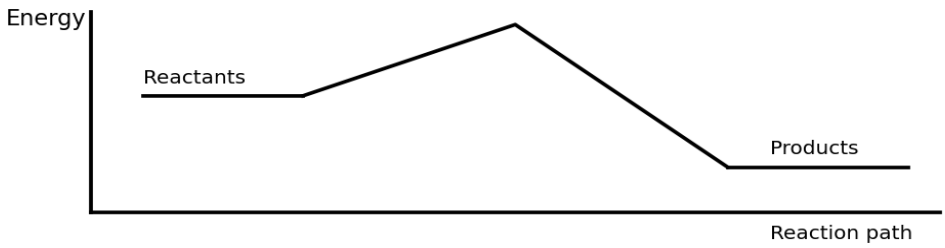
4. Aqueous copper(II) sulphate is electrolysed using inert carbon electrodes. What is deposited at the cathode?



A. Copper
B. Oxygen
C. Sulphur
D. Hydrogen only

Correct answer: A
Explanation: Cu^{2}■ ions are preferentially discharged at the cathode: Cu^{2}■ + 2e■ → Cu(s).

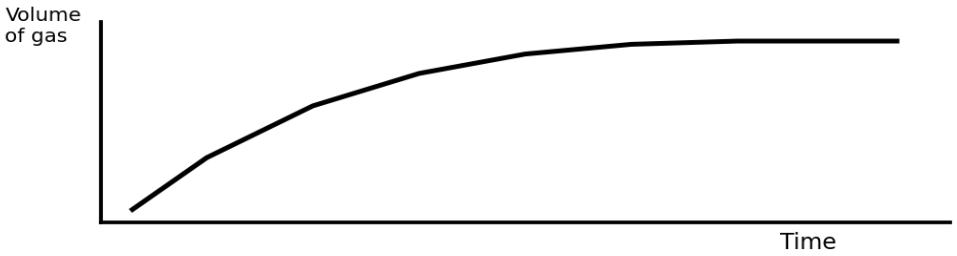
5. The energy profile shown represents a reaction in which the products have



A. higher energy than reactants
B. the same energy as reactants
C. lower energy than reactants
D. zero activation energy

Correct answer: C
Explanation: Products lie at a lower energy level than reactants, so the reaction is exothermic.

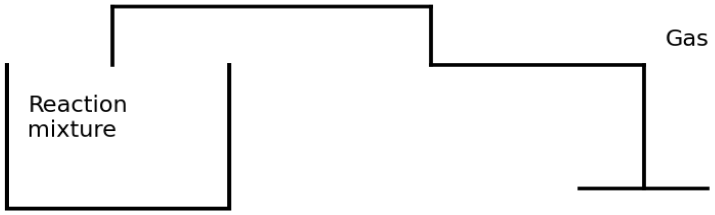
6. The graph shows the volume of gas produced against time. At the point where the curve becomes horizontal, the reaction



A. has reached completion
B. is at its fastest rate
C. has just started
D. has reversed completely

Correct answer: A
Explanation: The volume no longer increases, indicating that no more gas is being produced and the reaction has effectively stopped.

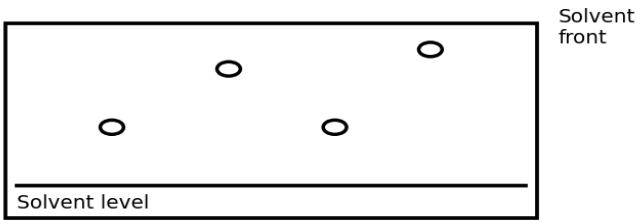
7. The gas generated in the reaction vessel is led through a delivery tube. Which property makes collection over water unsuitable for ammonia?



A. It is colourless
B. It is less dense than air
C. It is highly soluble in water
D. It has a pungent smell

Correct answer: C
Explanation: Ammonia dissolves readily in water, so it cannot be collected effectively by displacement of water.

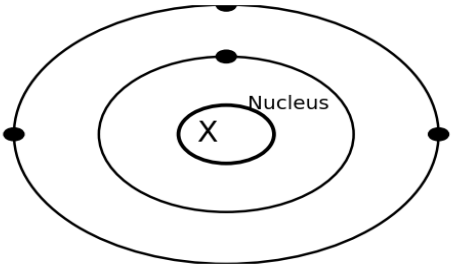
8. In paper chromatography, the spots shown separate because the components



A. have identical solubilities
B. move at different rates between the phases
C. all evaporate at once
D. react with the paper

Correct answer: B
Explanation: Different attractions to the mobile solvent and stationary paper cause different migration rates.

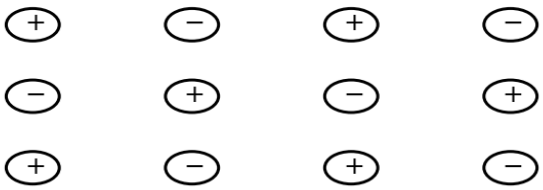
9. The atom represented has an outermost electron shell containing how many electrons if its arrangement is 2,8,1?



A. 1
B. 2
C. 8
D. 11

Correct answer: A
Explanation: The third shell contains one electron; the atom has electronic configuration 2,8,1.

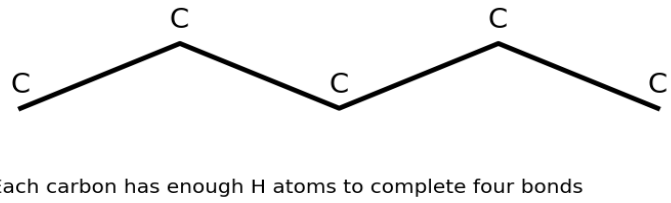
10. The alternating positive and negative ions in the lattice are held together mainly by



A. metallic bonding
B. hydrogen bonding
C. electrostatic attraction
D. van der Waals forces

Correct answer: C
Explanation: An ionic crystal is held together by strong electrostatic attraction between oppositely charged ions.

11. The carbon chain shown contains five carbon atoms joined by single bonds. The parent hydrocarbon is

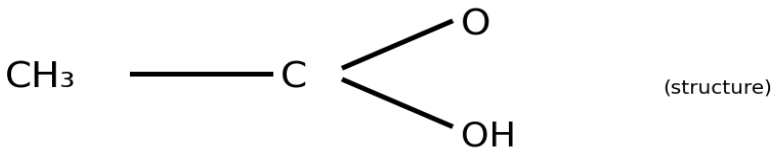


Each carbon has enough H atoms to complete four bonds

A. pentane
B. pentene
C. pentyne
D. pentanol

Correct answer: A
Explanation: A five-carbon saturated open-chain hydrocarbon is named pentane; the suffix -ane indicates single bonds.

12. The functional group shown is characteristic of a

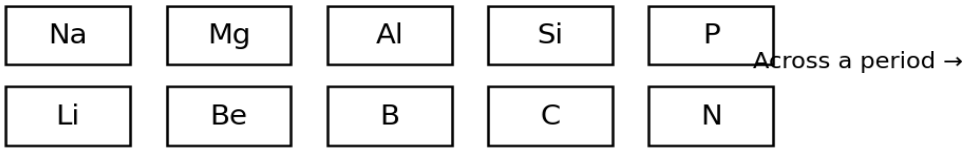


(structure)

A. ketone
B. carboxylic acid
C. ether
D. aldehyde

Correct answer: B
Explanation: The –COOH group is the carboxyl functional group found in carboxylic acids.

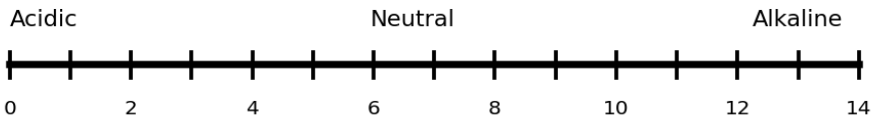
13. Across a period from left to right, the general trend in atomic radius is that it



A. increases
B. decreases
C. remains constant
D. first doubles

Correct answer: B
Explanation: Nuclear charge increases while added electrons enter the same main shell, pulling the electron cloud closer.

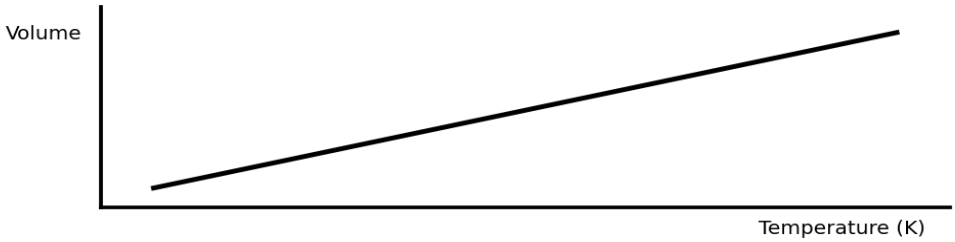
14. A solution with pH 3 is located in which region of the scale?



A. alkaline
B. neutral
C. acidic
D. strongly basic

Correct answer: C
Explanation: A pH below 7 indicates an acidic solution.

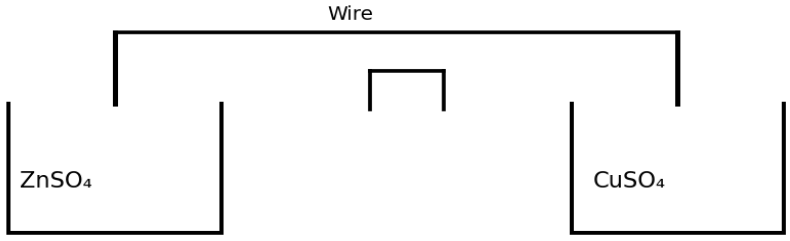
15. The straight-line graph shows gas volume against absolute temperature at constant pressure. It illustrates



A. Boyle's law
B. Charles's law
C. Graham's law
D. Dalton's law

Correct answer: B
Explanation: At constant pressure, gas volume is directly proportional to absolute temperature (Charles's law).

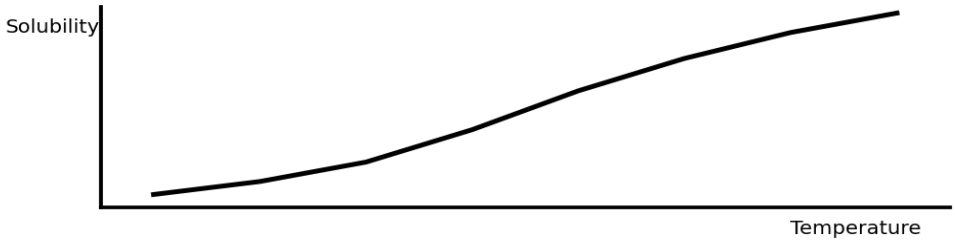
16. In the cell diagram, electrons flow through the external wire from the



A. copper electrode to zinc electrode
B. zinc electrode to copper electrode
C. salt bridge to zinc
D. electrolvte to the wire

Correct answer: B
Explanation: Zinc is oxidised and releases electrons; these travel through the wire to the copper electrode.

17. From the solubility curve, the solubility of the substance generally



A. decreases as temperature rises
B. increases as temperature rises
C. does not change
D. becomes zero at high temperature

Correct answer: B
Explanation: The upward-sloping curve indicates increasing solubility with increasing temperature.

18. Structures I and II have the same molecular formula but different structural arrangements. They are

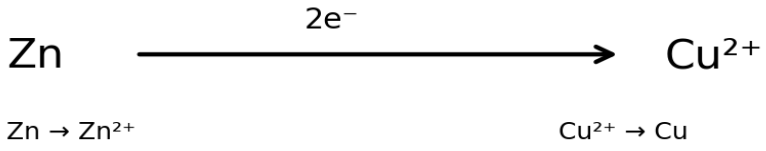
$CH_3—CH_2—CH_2—CH_3$ I

$CH_3—CH(CH_3)—CH_3$ II

A. isotopes
B. allotropes
C. structural isomers
D. homologues

Correct answer: C
Explanation: They are both C■H■■ but have different carbon skeletons, so they are structural isomers.

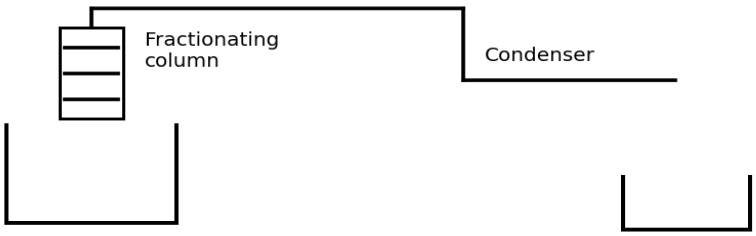
19. The diagram shows electron transfer from zinc to copper(II) ions. Zinc is therefore



A. reduced
B. oxidised
C. a catalyst
D. neutralised

Correct answer: B
Explanation: Zinc loses electrons to form Zn²■; loss of electrons is oxidation.

20. The fractionating column in the apparatus helps to separate liquid components because it



A. prevents boiling
B. provides repeated vaporisation and condensation
C. removes all impurities chemically
D. lowers the atmospheric pressure

Correct answer: B
Explanation: Repeated vaporisation and condensation improve separation of liquids with close boiling points.